

$$\int \frac{1}{(x+2)(x-2)} dx$$

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$$= \int \frac{A}{x+2} + \frac{B}{x-2}$$

$$= 1 = A(x-2) + B(x+2)$$

$$= 1 = A(-2-2) + B(-2+2)$$

$$= 1 = A(-4) + B(0)$$

$$= \frac{1}{-4} = \frac{-4A}{-4}$$

$$= A = -\frac{1}{4}$$

$$1 = A(2-2) + B(2+2)$$

$$1 = A(0) + B(4)$$

$$\frac{1}{4} = \frac{4B}{4}$$

$$B = \frac{1}{4}$$

$$\int \left(\frac{-1/4}{x+2} + \frac{1/4}{x-2} \right) dx$$

$$= -\frac{1}{4} \int \frac{1}{x+2} dx + \frac{1}{4} \int \frac{1}{x-2} dx$$

$$= -\frac{1}{4} \ln|x+2| + \frac{1}{4} \ln|x-2| + C$$

$$= \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + C$$